Where should USD 4.7B of SAF plant go in Kalimantan?

The user A SAF network planner choosing sites before engineering starts

Today A few hand-drawn layouts in a spreadsheet, with no proof of quality and no view of where a plan breaks

The cost of error A misplaced large refinery fixes USD 765M of capital and its haulage for 20 years

6,800 kt

biomass at 80 sites

1,200 kt

SAF, four airports

6,000 kt

biomass needed

20 years

CapEx life



One browser app, from inputs to an audited network

A

Map

All 80 sites, flows and costs, click any route

B

Engine

Rust solver with a proven bound

C

Simulator

Change any input, rerun for real

D

Decision support

Answers from solver records only

E

Evidence

Baselines, audit, report, failures



Hosted on GitHub Pages. No server, no account, no API key.

The engine proves how close it is to the best network

1 Bound

Lagrangian relaxation splits the model by site

2 Search

Branch-and-bound over refinery sites and sizes

3 Price

Exact min-cost flow for every design

4 Refine

Move plants to the geometric median, on land

Independent check on the same model

	Rust engine	HiGHS
Annual cost, USD M	890.32	890.32
Proven lower bound	883.60	866.74
Gap, time	0.75%, 60 s	2.65%, 480 s

Both solvers reached the same cost. On 5 smaller cases solved by HiGHS, the Rust bound was valid every time.

Six large refineries, inside the biomass clusters



USD 890.1M

per year, all costs

USD 742/t

SAF, before feedstock

23 depots

all at refineries

71 of 80

sites used

Measured gains, and where the plan breaks



Fails, and says why, when

+13.3% demand beyond feedstock

< 0.176 yield, inside the allowed 15 to 25%

5 plants or fewer cannot take 6,000 kt

The planner stays in control, and can check every number

Trust

No language model, so no invented numbers

Each answer lists the solver fields it read

What-if questions rerun the real solver

Questions outside the model are declined with a reason

A separate audit recomputes 11 checks from raw flows

Impact

USD 61M to 77M per year

saved against rule-of-thumb layouts, measured

Reality check

Official demand is about 12 times today's real jet use at these airports, so this is a hub-scale plan. Site moisture data is the next thing to test.